A-A-54079
23 February 1990

COMMERCIAL ITEM DESCRIPTION

SURVEYOR, DENTAL, CLASP, STATIONARY

The General Services Administration has authorized the use of this Commercial Item Description as a replacement to Federal Specification GG-S-005B, which is cancelled.

Salient characteristics:

This commercial item description covers the following styles of stationary dental clasp surveyors suitable for use in designing dental prosthesis.

Style 1 - Surveyor with spindle spring

Style 2 - Surveyor without spindle spring

Description. The dental clasp surveyor is intended for use in designing and constructing removable partial dentures. The surveyor shall consist essentially of a horizontal surveyor base, an upright post and crossarm with a spindle bearing or a surveying arm with a spindle bearing, a vertical spindle, a spindle locking screw, a tool holder and storage compartment, a swivel base surveying table, and other accessories.

Materials. The dental surveyor shall be fabricated of the materials shown below for style 1 and style 2 surveyors.

Design and dimensions. The dental clasp surveyor shall be furnished as style 1 or style 2 and shall conform to Defense Personnel Support Center drawing 24048 for design and dimensions and the requirements specified herein.

Main base. The main base shall be an accurately machined horizontal surface on which the surveying table moves freely. The bottom of the surveyor base shall have skid-resistant rubber feet (style 1) or a skid-resistant rubber rim (style 2).

Surveying arm or upright post and crossarm. A one-piece surveying arm (style 1), or a crossarm extending from a vertical upright post (style 2) shall remain fixed in position and shall house a spindle bearing at its free end.

AMSC N/A FSC 6520

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Materials (style 1)

Material	Surveyor Components
Cast aluminum	Main base, surveying table (swivel base, swivel base mount, movable clamping jaw), surveying arm.
Aluminum	Spindle knob, movable jaw thumbscrew, tool storage compartment, swivel base locking screw knob.
Stainless steel	Chuck thumbscrew, spindle, swivel base locking screw, spindle spring, carbon rod sheath, undercut gauge - .030 inch
Steel; gold anodized	Undercut gauge - .010 inch
Steel; gray oxide	Undercut gauge - .020 inch
Steel; black oxide	Blocking out tool
Steel; carburized and hardened	Wax trimmer
Nickel-plated brass	2 degree and 6 degree taper tools.
Nickel-plated steel	Surveying rod.
Tempered steel	Swivel base locking clamps, swivel base retaining ring.
Brass	Swivel base locking screw liner, surveying table ball and socket joint.
Teflon composition	Spindle bearings.
Carbon	Spindle scribing rods.
Rubber	Surveyor base feet.
Nylon	Spindle friction button.

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Materials (Style 2)

Material	Surveyor Components
Cast aluminum	Main base, surveying table (swivel base, swivel base mount, movable clamping jaw).
Cast aluminum w/steel (carburized and hardened) sleeve bearing	Crossarm, spindle bearing.
Aluminum	Upright column, column cap, movable jaw thumbscrew, tool storage compartment and cap.
Aluminum w/spring loaded brass insert	Spindle locking screw.
Stainless steel	Spindle, analyzing rod, undercut gauge .010 inch, swivel base locking screw.
Stainless steel copper plated	Undercut gauge -.020 inch.
Stainless steel black oxide	Undercut gauge -.030 inch.
Steel; carburized and hardened	Wax trimmer.
Stainless steel	Carbon sheath.
Brass	Ball pivot, ball retaining ring.
Carbon	Round carbon marker.
Rubber	Surveyor base feet.

Spindle. A spindle, when inserted through the bearing shall be perpendicular to the base. A spindle locking screw inserted through the end of the surveying arm or crossarm shall be used for locking the spindle at a particular height. The unlocked spindle shall rotate and move vertically in its bearing and shall always remain perpendicular to the top surface of the surveyor base. Style 1 surveyor shall be equipped with a spindle spring located below the spindle knob to return the spindle to a vertically high position when the spindle locking screw is disengaged.

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Tool holder. A precision V-way and self-adjusting tool holder and locking nut shall be located on the bottom end of the spindle to permit removing and installing tools in the chuck.

Tool storage compartment. The surveyor shall be equipped with a tool compartment for storage of working tools when not in use. There shall be holes drilled in the top surface of the surveyor to hold working tools during use.

Swivel base surveying table. The surveying table shall have an adjustable swivel base mounted on a ball-and-socket joint. The table shall include two stationary clamping jaws, a movable clamping jaw, a movable jaw thumbscrew, a swivel base mount, and a swivel base locking screw.

Accessories. Style 1 surveyor shall be furnished with one surveying rod, one cylindrical blocking-out tool, one 6° taper tool for use on short teeth, one 2° taper tool for use on long teeth, three undercut gauges, (one each, 0.010, 0.020, and 0.030 inch), one carbon rod sheath and one set of six carbon markers. Style 2 surveyor shall be furnished with one analyzing rod, three undercut gauges (one each, 0.010, 0.020, and 0.030 inch), one wax trimmer, one carbon sheath, and one dozen round carbon markers.

Runout. The surveying spindle shall be perpendicular to the top surface of the surveyor base and shall have a runout not exceeding 0.005 inch when tested by an accurate method.

Finish. All surfaces, except for plated hardware components and those listed below, shall have a machined, black painted surface (style 1) or a black, textured enamel finish (style 2). The working surface of the main base and upright column (style 2) shall be machined. The surveying spindle shall be machined and centerless ground to produce a straight, concentric rod. The top surface of the swivel base shall have a smooth finish. The outer surface of all screws and knobs shall be machined with a knurled finish.

Instructions. Adequate and accurate operating instructions shall be furnished with each surveyor.

Marking. Each surveyor shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

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Workmanship. The surveyor shall be free from burrs, nicks, rough edges, or other defects which may detract from its appearance or impair its serviceability.

Unit. Each (EA). One surveyor, as specified, constitutes one unit.

Unit package. Each surveyor shall be disassembled as far as is practicable. The components comprising the surveyor shall be wrapped in neutral tissue or other suitable material and suitably packaged to prevent breakage during shipment. As an alternate, each unit may be shrink-wrapped or blister-packed with clear plastic to a sheet of corrugated fiberboard. The vacuum packaged sheet shall be folded into a box which encloses the article being packed. Closure shall be effected by use of tape at all seams.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one surveyor. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, and sampling for tests (when applicable) shall be conducted at inspection level S-1. For tests, acceptance number shall be zero (0) for all sample sizes.

Examination. Examination shall include, but not be limited to, the following:

Defects

Top surface of base or bottom surface of surveying table swivel base mount not smooth or level.
Base, surveying arm, upright column or surveying table has significant casting defects.
Burrs, nicks or feathers present on surveyor components.
Surveyor components not properly assembled.
Collet-chuck threads defective or missing.
Exposed surfaces not free from rough edges.
Spindle not centerless ground.
Surveyor components not properly assembled.
Component surfaces not clean or showing corrosion.
Plating of components shows signs of peeling or flaking.
Enamel finish not as specified.
Instrument identification missing.
Instrument identification not as specified.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquistion Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-558-2700	SURVEYOR, DENTAL, CLASP, STATIONARY

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

PROJECT NO. 6520-2063

LOCATION: NCR Disk No. 7

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54079	Surveyor, Dental, Clasp, Stationary

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last First MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City State ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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